Microplastic Detection in Water

Microscopy analysis report - YOLOv8n - Baseline Fragment Detector

Analysis: 2026-11-03T08:04:59.173325+00:00 | ID: 3842fc7b304f40d390fe426d6403db6b

Input: test_sample_01.jpg

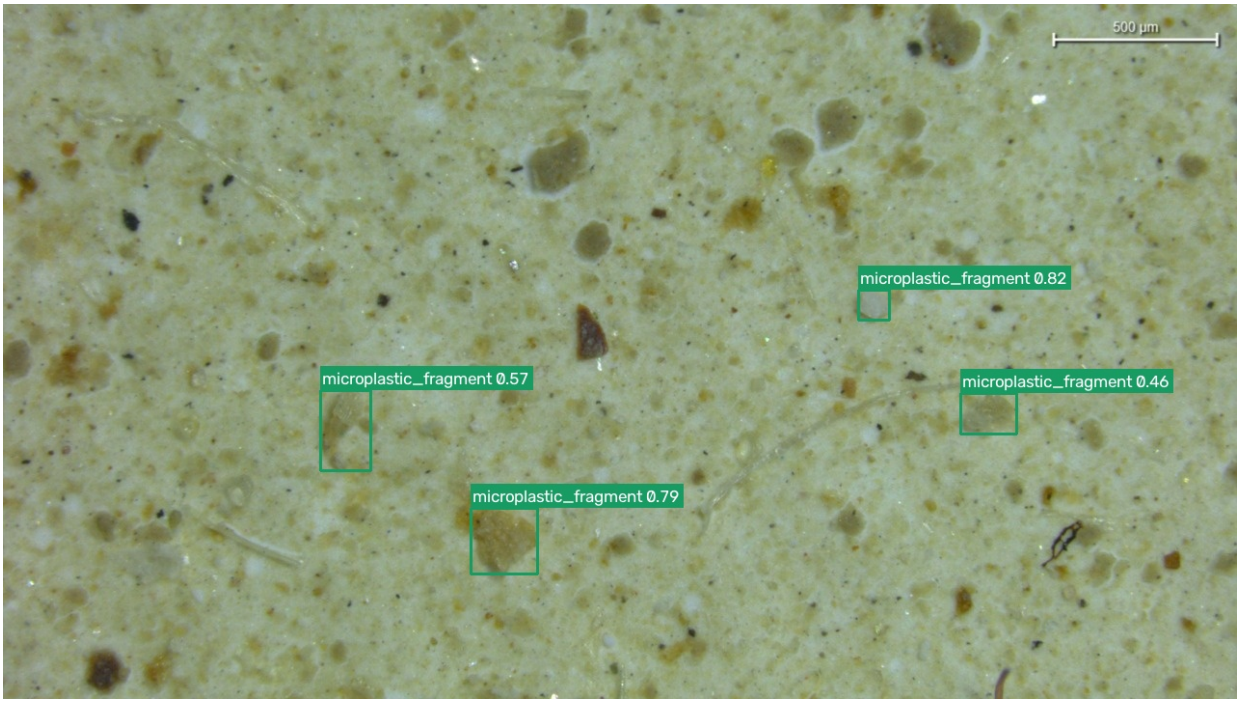
Model: YOLOv8n - Baseline Fragment Detector | Version: baseline

Particles: 4 | Mean confidence: 66.1% | Image count band: Low

Class counts: microplastic_fragment: 4

Inference: 220.6 ms | Threshold: 0.25

Calibration: Not provided; dimensions are in pixels.



Results are image-based detections of suspected particles. They do not independently confirm that a particle is plastic or identify its polymer composition. Confirmatory chemical analysis and a validated sampling protocol may be required for scientific conclusions.

Count bands: Low < 10, Moderate 10-29, High >= 30. These thresholds are configurable demonstrations.

Scientific Limitations

- Morphology detection is not polymer identification.
- Visible-light images may contain sediment, bubbles, and natural fibers.
- Performance may vary between microscopes, lighting, and sample matrices.
- Pixel measurements are not physical measurements without calibration.
- Particle count per image is not concentration per litre.

Detected particles

ID	Morphology Class	Confidence	Width (px)	Height (px)	Area (px²)
1	microplastic fragment	82.5%	31.8	31.1	990.9
2	microplastic fragment	78.9%	68.4	68.1	4655.9
3	microplastic fragment	57.0%	52.1	82.8	4312.1
4	microplastic fragment	46.0%	58.7	41.7	2445.5